AWS VPC PEERING PROJECT

Project Report

Project Title: Secure Communication Between Two AWS VPCs Using VPC Peering

Cloud Platform: Amazon Web Services (AWS)

AWS Region: Asia Pacific (Sydney) — ap-southeast-2

Project Type: Cloud Networking

Technologies Used:

- Amazon VPC
- VPC Peering
- Amazon EC2
- Route Tables
- Security Groups
- Amazon Linux 2023

Project Objective: To establish and verify secure network communication between two separate Amazon VPCs using VPC Peering.

1. Introduction

Amazon Web Services (AWS) is a cloud computing platform that provides services for building and managing secure, scalable, and reliable infrastructure.

Amazon Virtual Private Cloud (Amazon VPC) allows users to create isolated virtual networks in the AWS cloud. VPC Peering enables two VPCs to communicate privately using their private IP addresses without requiring traffic to travel over the public internet.

In this project, two separate VPCs were created in the Asia Pacific (Sydney) region. A VPC Peering connection was established between them, and route tables and security groups were configured to enable controlled communication between EC2 instances in both VPCs.

The connectivity was verified by sending ICMP ping requests between the application instances in both directions.

2. Project Objectives

The main objectives of this project were:

- To create two separate Amazon VPCs with different IPv4 CIDR blocks.
- To create and configure the required subnets.
- To establish a VPC Peering connection between the two VPCs.
- To configure route tables for private communication between the VPCs.
- To launch EC2 instances and configure security groups.
- To use a bastion host to access private EC2 instances.
- To test communication between the application instances using ICMP ping.
- To verify that security group rules control network connectivity.

3. AWS Network Architecture and Configuration

3.1 VPC Configuration

Two Virtual Private Clouds were created in the AWS Asia Pacific (Sydney) region.

Configuration	VPC A	VPC B
VPC Name	ewb-vpc-a	ewb-vpc-b
IPv4 CIDR	10.10.0.0/16	10.20.0.0/16
Private Subnet	ewb-a-private	ewb-b-private
Private Subnet CIDR	10.10.2.0/24	10.20.1.0/24

VPC A also contains a public subnet named ewb-a-public with CIDR 10.10.1.0/24. An Internet Gateway was attached to VPC A to provide internet connectivity for the public subnet.

3.2 VPC Peering Connection

A VPC Peering connection named ewb-a-b-peer was created between VPC A and VPC B.

The connection was accepted successfully, and its status became **Active**. The peering connection allows private communication between resources in the two VPCs when the appropriate routing and security rules are configured.

3.3 EC2 Instances

Three Amazon EC2 instances were launched using Amazon Linux 2023 and the t3.micro instance type.

Instance Name	Private IPv4 Address	Purpose
ewb-bastion	10.10.1.10	Bastion host for accessing private instances
ewb-app-a	10.10.2.10	Application instance in VPC A
ewb-app-b	10.20.1.10	Application instance in VPC B

The bastion host is located in the public subnet of VPC A. The application instances are located in private subnets and are accessed through the bastion host.

4. Routing and Security Configuration

4.1 Route Table Configuration

Route tables were configured to allow traffic between VPC A and VPC B through the VPC Peering connection.

The following routes were configured:

Route Table	Destination CIDR	Target
ewb-rt-a-private	10.20.0.0/16	ewb-a-b-peer
ewb-rt-b-private	10.10.0.0/16	ewb-a-b-peer
ewb-rt-a-public	10.20.0.0/16	ewb-a-b-peer
ewb-rt-a-public	0.0.0.0/0	Internet Gateway

The peering routes allow resources in each VPC to reach the other VPC's CIDR range. The Internet Gateway route provides internet routing for the public subnet.

4.2 Security Group Configuration

Security groups were configured to control traffic to the EC2 instances.

Bastion Host — ewb-sg-bastion

- SSH traffic on TCP port 22 was allowed from the user's public IP address.
- Access was used to connect to the private application instances through the bastion host.

Application A — ewb-sg-app-a

- SSH access was allowed from the bastion host at 10.10.1.10/32.
- ICMP traffic was allowed from Application B at 10.20.1.10/32.

Application B — ewb-sg-app-b

- SSH access was allowed from the bastion host at 10.10.1.10/32.
- ICMP traffic was allowed from the bastion host at 10
-
- .10.1.10/32.
- ICMP traffic was allowed from Application A at 10.10.2.10/32.

These rules enabled the required connectivity while restricting access to specified source addresses.

5. Testing and Results

5.1 SSH Connectivity

The bastion host was used to establish SSH connections to the private EC2 instances in VPC A and VPC B. This verified that the private instances could be accessed through the bastion host.

5.2 Connectivity from Application A to Application B

A ping test was executed from Application A (10.10.2.10) to Application B (10.20.1.10).

Command:

ping -c 4 10.20.1.10

Result: **Successful.** Four packets were transmitted and four packets were received, with 0% packet loss.

5.3 Connectivity from Application B to Application A

A ping test was executed from Application B (10.20.1.10) to Application A (10.10.2.10).

Command:

ping -c 4 10.10.2.10

Result: **Successful.** Four packets were transmitted and four packets were received, with 0% packet loss.

5.4 Security Group Verification

The inbound ICMP rule allowing traffic from Application A was temporarily removed from Application B's security group. The ping test then failed, demonstrating that the security group rule controlled connectivity.

After the ICMP rule was restored, the ping test succeeded again.

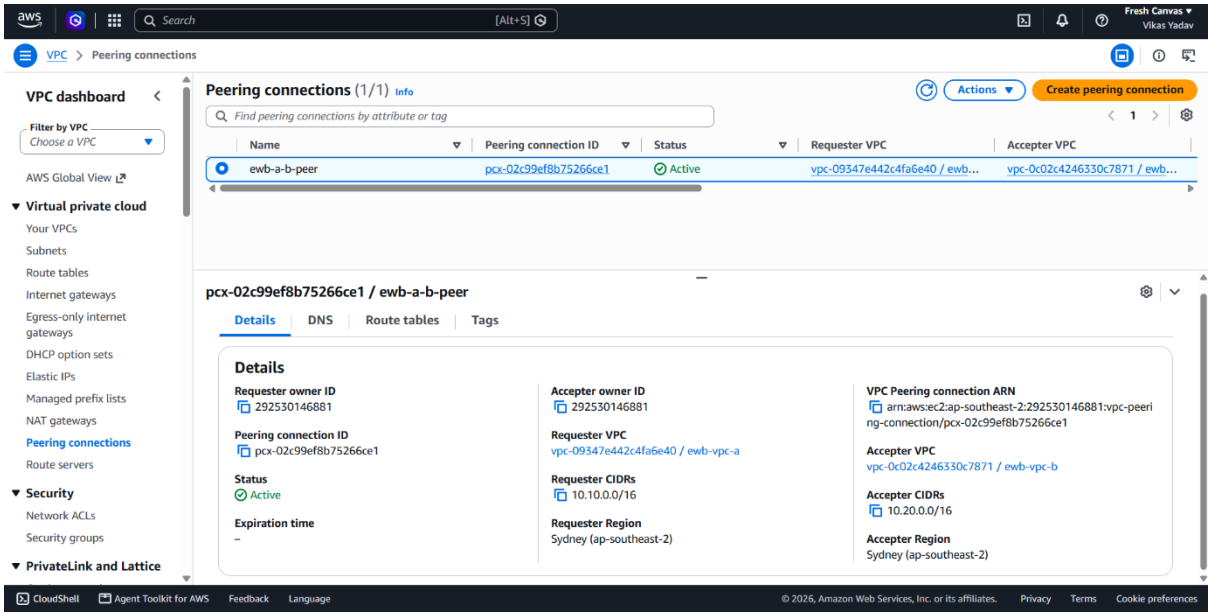
5.5 Summary of Results

Test	Result
VPC Peering status	Active
SSH access through bastion	Successful
Application A to Application B ping	Successful
Application B to Application A ping	Successful
Security group restriction test	Verified

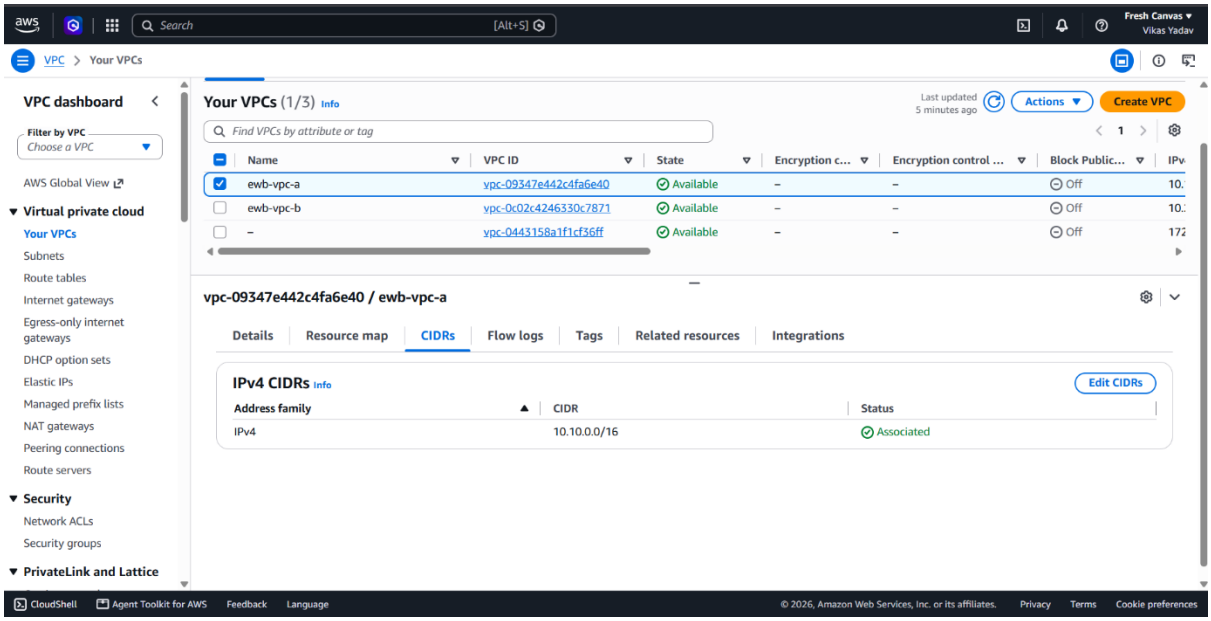
The results demonstrate that private communication between the two VPCs was successfully established and that security group rules controlled the allowed traffic.

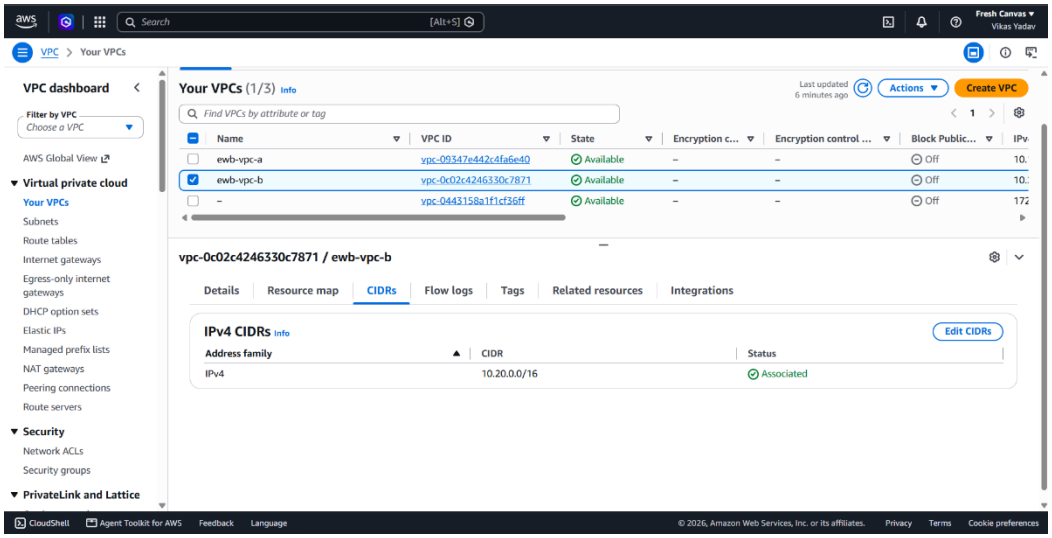
6. Project Evidence

6.1 VPC Peering Connection Status

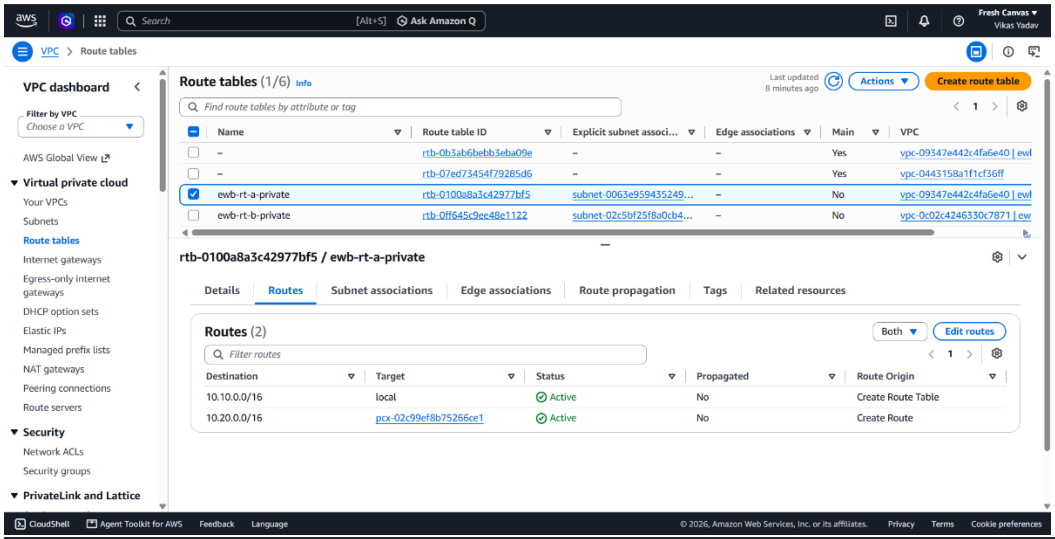


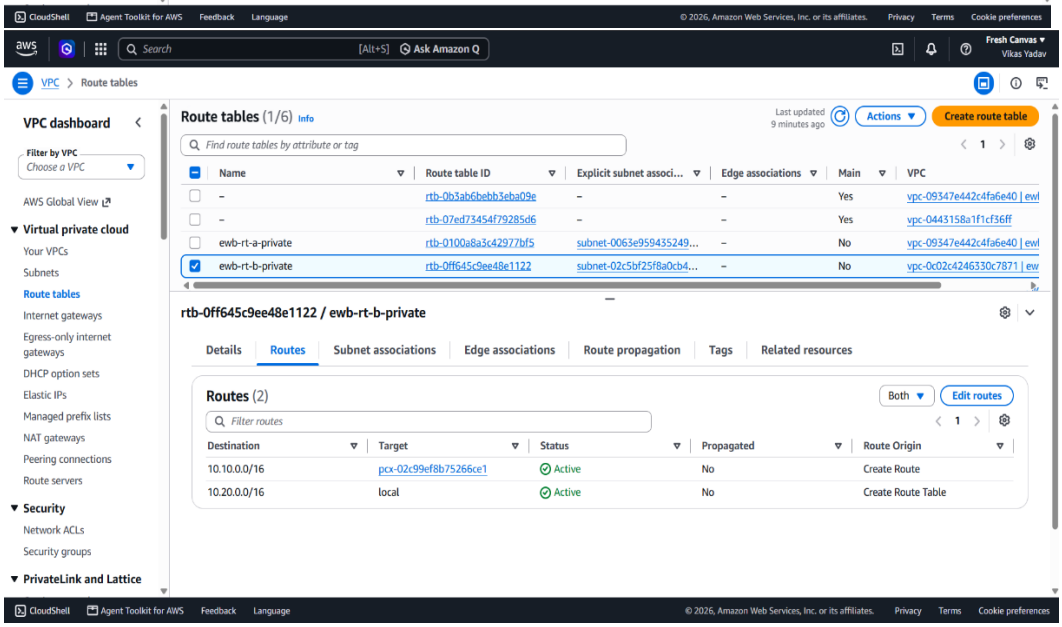
6.2 VPC Configuration



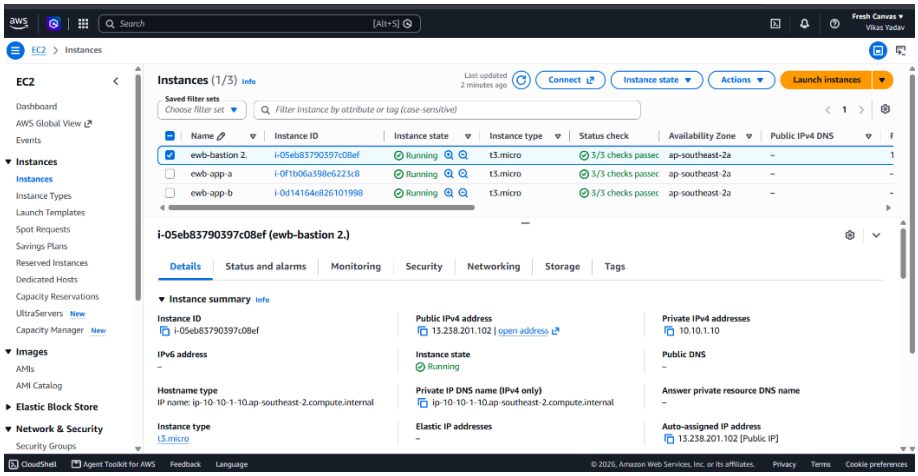


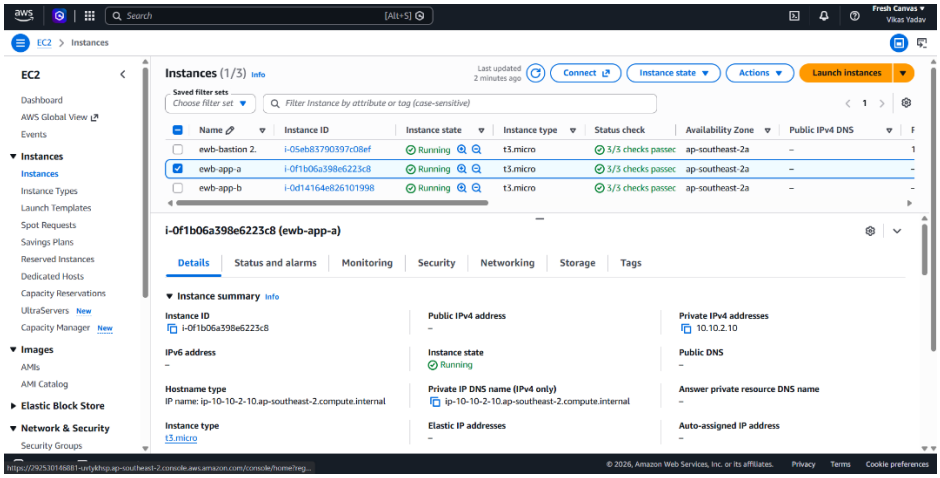
6.3 VPC A Private Route Table

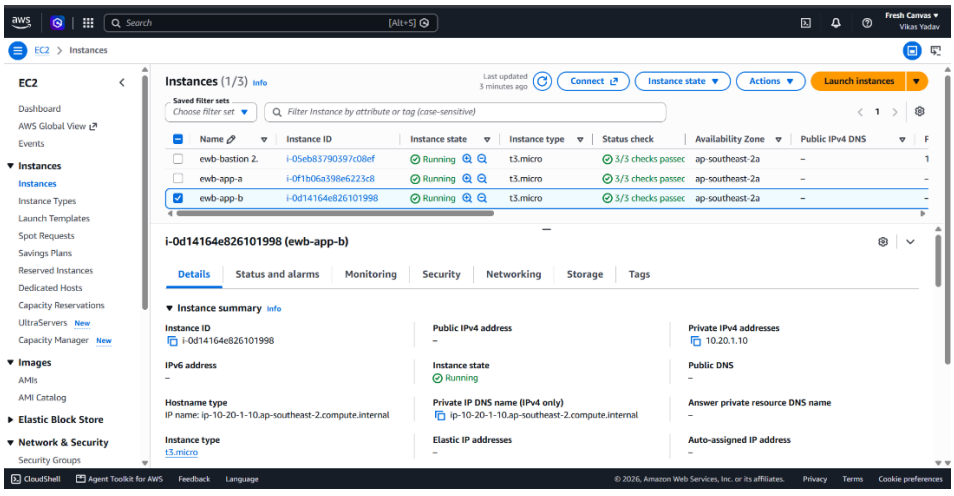




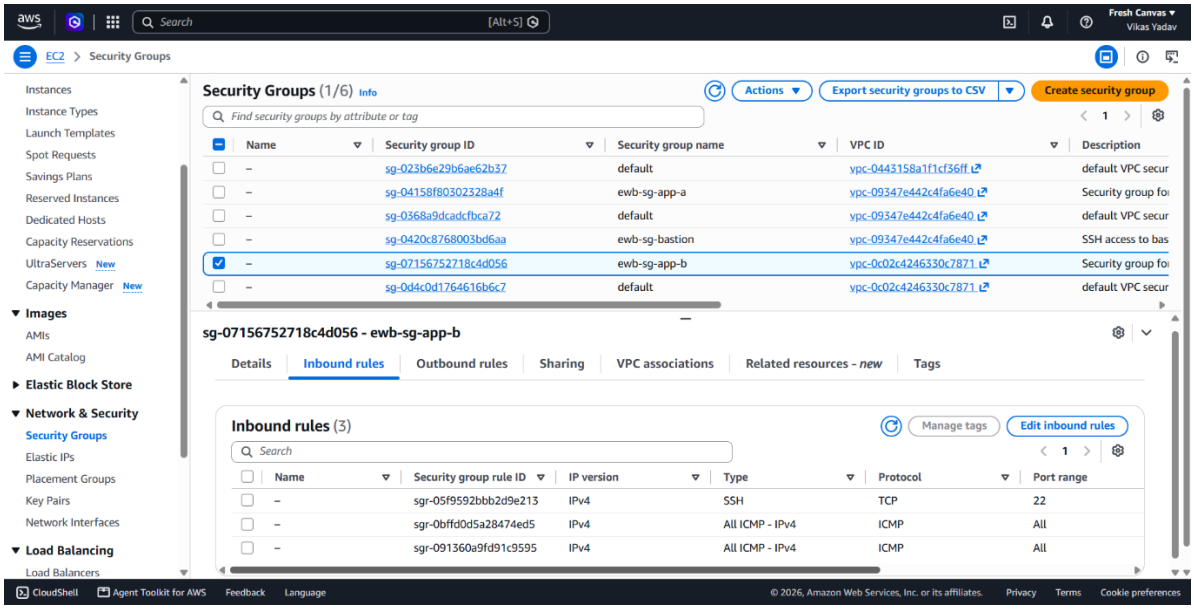
6.5 EC2 Instances



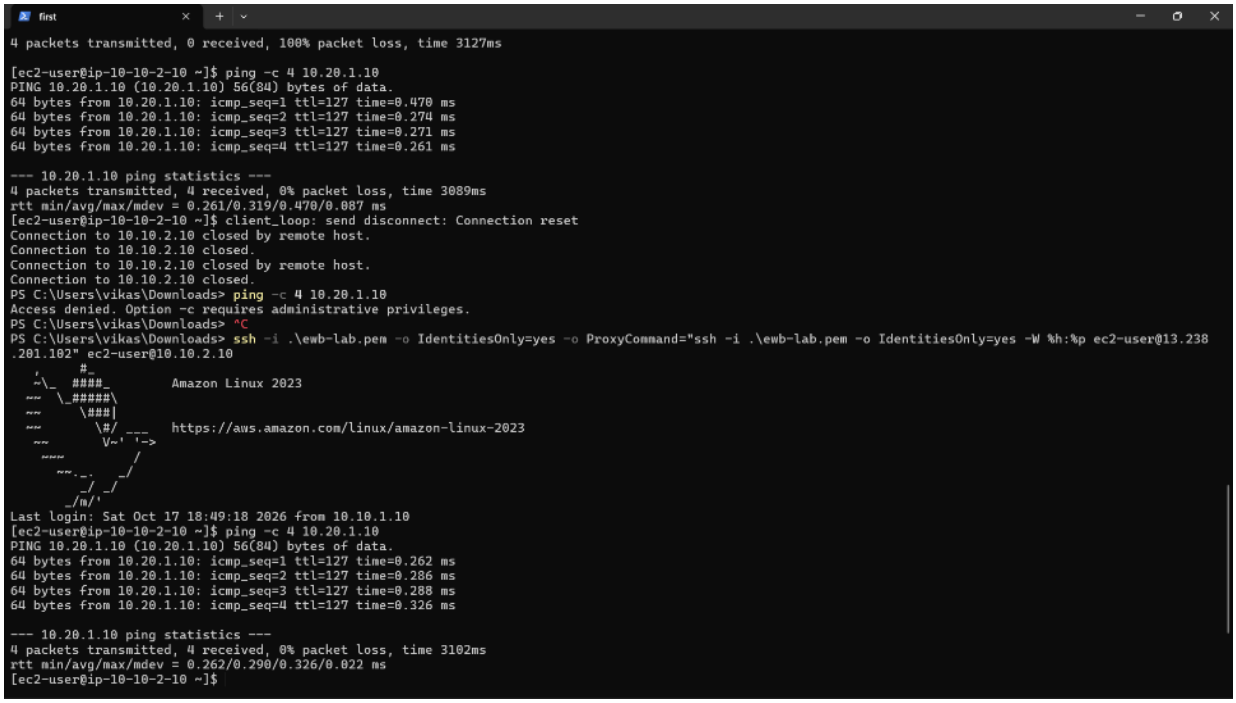




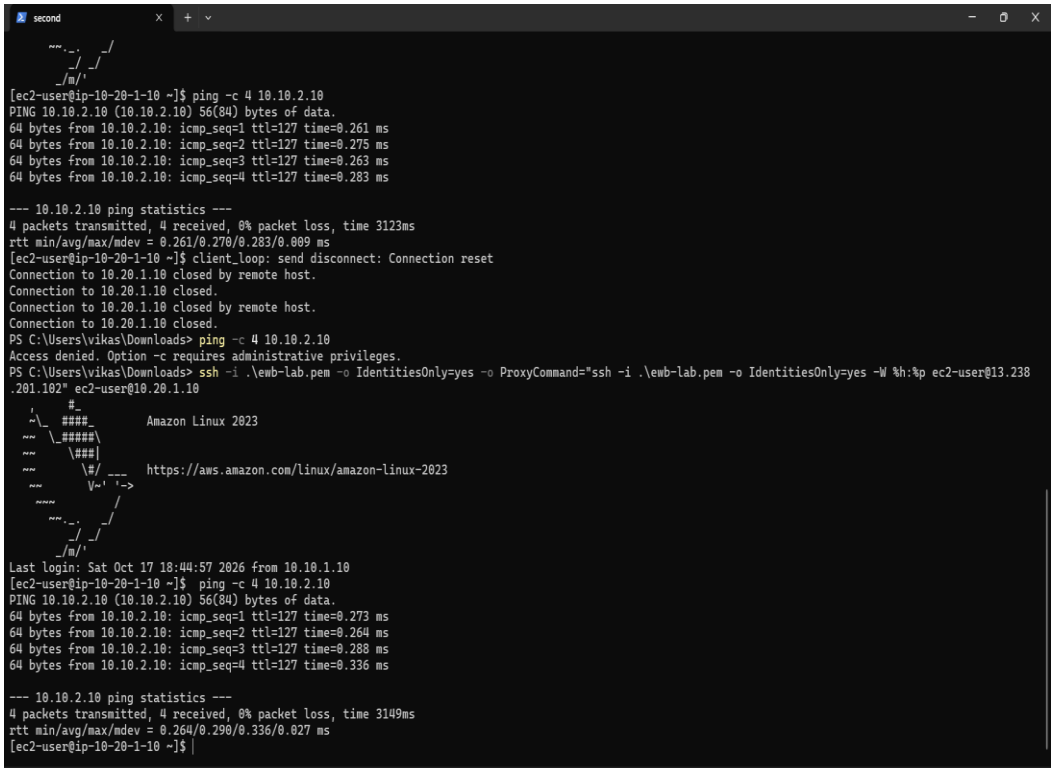
6.6 Application B Security Group Rules



6.7 Connectivity Test: Application A to Application B



6.8 Connectivity Test: Application B to Application A



7. Conclusion

The AWS VPC Peering project was successfully implemented using Amazon Web Services. Two separate VPCs were created with different CIDR blocks, and a VPC Peering connection was established between them.

Route tables were configured to direct traffic between the VPCs, and security groups were used to control access to the EC2 instances. A bastion host provided SSH access to the private application instances.

Connectivity testing confirmed successful communication between Application A and Application B in both directions, with zero packet loss. The security group verification test also demonstrated that network access depended on the configured inbound rules.

This project provided practical experience with AWS networking, VPC Peering, route tables, EC2 instances, security groups, SSH access, and network troubleshooting.

Overall, the project demonstrated how to establish and verify controlled private communication between two AWS VPCs.